Lab 4: Working with Vectors and Implementing the Big Five

Name: _____
Date: _____
Class/Section: _____

Objective

1. Understand and use std::vector from the C++ STL.
2. Implement the Big Five in a custom vector class (MyVector).
3. Compare performance of std::vector and MyVector for searching, sorting, copying, and moving.

Task 1 & 2: Performance Comparison

Instructions

1. Create std::vector<int> and MyVector<int> objects of sizes: 1,000, 10,000, and 100,000.
2. Populate vectors with integer values.
3. Implement operations for both vector types:
    - Search (linear and binary)
    - Sort
    - Copy
    - Move
4. Measure time taken for each operation.

Performance Comparison Table

Operation	Vector Type	n=1000	n=10000	n=100000
Linear Search	std::vector			
Linear Search	MyVector			
Binary Search	std::vector			
Binary Search	MyVector			
Sort	std::vector			
Sort	MyVector			
Copy	std::vector			

Copy	MyVector			
Move	std::vector			
Move	MyVector			

Questions

1. How does the performance of MyVector compare with std::vector for searching, sorting, copying, and moving?
2. How does vector size affect performance?
3. Why are move constructors and move assignment operators beneficial for performance?
4. How does using a template improve MyVector's reusability?

Submission Guidelines

- Submit all code files.
- Submit a performance analysis report with the filled comparison table.
- Submit answers to all questions.